Algoma University
Faculty of Computer Science & Technology

COSC 2307
Database Programming
Winter 26

1. Instructor Information

Instructor: Dr. Mahreen Nasir
Office: N/A
Office Phone: TBA
E-mail: mahreen.nasir@algomau.ca

*Note: Only email originating from a valid Algoma university student account will be accepted from students wishing to contact the instructor. Please include your full name, student ID and related course section in your correspondence. Do not spam with multiple or lengthy emails. Kindly expect a maximum of **48-72** hrs response time (excluding weekends).*

2. Course Information

Course Calendar Description:
Students are introduced to database programming using SQL. The topics covered include data definition, single and multiple table queries, updates, database administration, embedded SQL, report formatting, data protection and security.

Prerequisite: COSC 1046 or equivalent programming experience. Students may not retain credit for both COSC 2926 E and COSC 2307 E.

Note: Students who have not obtained prerequisite credits should consider dropping the course, choosing to remain enrolled at their own risk. Course prerequisites are detailed on the Employee Portal Course Calendar and in the AU Academic Calendar.

3. Learning Outcomes:

By the end of this course, students will be able to:

- Demonstrate database knowledge and understanding of basic concepts of database.
- Demonstrate the knowledge and understanding of structured query language (SQL), in particular Data Definition Language (DDL) and Data Manipulation Language (DML).
- Construct queries using SQL.
- Design and analyze a commercial relational database system (for example Oracle) by writing SQL using the system.
- Demonstrate the knowledge, understanding and methodology of database design by translating requirements into a data model.
- Apply and use computer-based software to design and create database.
- Apply normalization to design a relational database.
- Create a GUI that uses SQL and Stored Procedures to access data in underlying tables.

4. Textbook(s):

- Database Concepts, 10th edition, David M. Kroenke, David J. Auer, Scott L. Vandenberg, Robert C. Yoder, published by Pearson (December 31, 2022) © 2024

** Students will be provided with all materials (open access textbooks), hence zero costs associated for the course materials.

ReferenceTextbook(s):

- Modern Database Management, Global Edition, Jeff Hoffer ,Ramesh Venkataraman , Heikki Topi ,2020, PEARSON, ISBN-10 : 1292263350, ISBN13.
- Learning SQL, A Step-By-Step Guide Using Oracle, Richard Earp and Sikha Bagui, 2003, Addison-Wesley, ISBN: 0-201-77363-5.

Additional Readings: Will be made available on Course site on Brightspace.

Additional Supplies, Materials or Equipment: Will work with different Database Management Systems including MS Access, SQLite and/or Oracle. More information on course Brightspace page.

5. Course Requirements and Assessment

Information on course requirements and assessments.

Assessment	Date of Evaluation (if known)	Weighting
Assignments (5 * 10%)	*See below the tentative schedule	50%
Midterm Exam		15%
Final Exam (Online Asynchronous)		35%
Total		100%

5.1. Assignments:

Over the course, five assignments will be delivered (based on the usage of different DBMS's), and each will be worth 10% towards the final grade. The assignments should be finished individually. All students are expected to submit their assignments by 11:59pm on the due date.

5.2. Late Submissions:

If an assignment or project report is submitted late, a penalty of 10 percent of that submission's grade will be assessed for each day it is late. If a submission is **late for over 3 days**, it will not be graded and therefore you will get **Zero**. **No deadline extension request will be considered** since the deadlines are known well ahead. However, in an emergency situation beyond students' control (e.g., medical etc), situation can be considered on an individual basis provided that doctors' note is submitted.

5.3. Midterm Exam:

There will be one midterm exam and it will be worth 15% towards the final grade. This exam will cover topics presented in all lectures that have been delivered before the exam date. Regularly check your LMS for announcements.

5.4. Final Exam:

The final exam will be comprehensive, covering all material in the course. It will contribute 35% towards your final grade.
The Final Examination Policy can be found at:
https://algomau.ca/wp-content/uploads/2018/10/Final-Examination-Policy.pdf

Note: Online proctoring tool "ProctorU will be used for mid term and final examination.Students must check regular announcements on course LMS (Brightspace) and follow details to book their exams via ProctorU.

5.5. Missed Tests and Exams:

Make up examinations for mid term and final term will ***not*** be conducted.

The Final Examination Policy can be found at:
http://algomau.ca/media/styleassets/pdf/final_examination_policy.pdf

https://drive.google.com/file/d/10QwzAFVXXCqXIrPdYua5puqwT1xJYUyM/view

6. Class Schedule (Tentative)

***Note: Students must regularly check course LMS and their emails for course related announcements , assignments submission and examination deadline calender.**

Week	Topic, activity, assignment, etc.	Textbook Chapter or Readings
1	Review of course outline, Structure and Policies **Part 1: Database Fundamentals**	**Chapter 1a & 1b**
2	**Part 1: Database Fundamentals**	**Chapter 2**
3	**Part 1: Database Fundamentals**	**Chapter 2**
4	**Part 1: Database Fundamentals**	**Chapter 3**
5	**Part 1: Database Fundamentals**	**Chapter 3**
6	**Part 2: Database Design**	**Chapter 4**

7	**Part 2: Database Design**	**Chapter 4**
8	**Part 2: Database Design**	**Chapter 5**
9	**Part 2: Database Design**	**Chapter 5**
10	**Part 3: Database Management**	**Chapter 6**
11	**Part 3: Database Management**	
12	Review and Discussions	

7. Important Dates:

Students must regularly check the university website for the academic calender and important dates.

https://drive.google.com/file/d/116qSe35Ew3WMOUSULNTRU_d84nwofFvC/view?usp=drive_link

8. Academic Dishonesty:

The University takes a very serious view of such offences as plagiarism, cheating, and impersonation. Penalties for dealing with such offences will be strictly enforced.

The following web site contains a complete policy statement on academic dishonesty and attendance. Students are encouraged to read this policy for further clarification of these issues:
https://algomau.ca/academics/academic-integrity-office/academic-integrity-for-students/support-and-resources/

9. Use of Online Exam Proctoring Software:

To ensure academic integrity during an online examination, "*an **online proctoring Proctor U tool** will be used".*

10. Attendance Policy:

The general regulations of the University require punctual and regular attendance at the various academic exercises. If there are extenuating circumstances related to an absence, the instructor should be notified. Absences in excess of 20% may jeopardize receipt of credit for the course.

11. Disability Accommodation:

If you are a student with a physical, learning, and/or psychological disability and plan to request any academic accommodations for this class, you are required to bring in an authorization letter from Disability Services listing the permitted accommodations. I will work with you to arrange your accommodations from the point i

time that you deliver and discuss such an authorization letter with me. The Coordinator of Disability Services will keep your disability documentation confidential. Contact information is as follows: Coordinator of Disability Services 705-949-2301 ext. 4221; learning@algomau.ca